Christopher P. McCormick

M.S. Aerospace Engineering

University of California, Los Angeles

Phone: +1 (949) 444-9470

Email: chris.mccormick.aero@gmail.com

Website: chrismccormick45.github.io

LinkedIn

CV

University of California, Los Angeles (UCLA)

M.S. Aerospace Engineering GPA: 3.81 / 4.00 Graduated: **Fall 2025**

Thesis: Data-Driven Flight Dynamics Modeling for a Tailless Unmanned Aerial System (advisor: Kunihiko Taira)

B.S. Aerospace Engineering GPA: 3.98 / 4.00, *Summa Cum Laude* Graduated: **Fall 2024**

PROJECTS & EXPERIENCE

- **Undergraduate & Graduate Student Researcher | UCLA Taira Lab** **Jun '23 – Dec '25**
  - Developed an optimization framework to accurately model the flight dynamics of the Dihedral-90 (D90) tailless UAS in collaboration with Prof. David Williams and PhD candidate Sai Simon at the Illinois Institute of Technology.
  - Applied Markov Chain Monte Carlo (MCMC) sampling through the Delayed Rejection Adaptive Metropolis (DRAM) algorithm to learn a set of stability derivatives that model dominant dynamics.
  - Validated models using five-fold cross-validation and performed uncertainty quantification through credible and prediction intervals to assess model accuracy.
  - Analyzed parameter distributions to assess model and parameter sensitivity, and confidence in results.
  - Processed and performed all analysis on MATLAB, using Simulink to model the control systems on board D90 and to simulate flight dynamics.
  - Thoroughly described this research and the results in my Master's Thesis, seen **here**.
- **Aircraft & Spacecraft Design Projects** **Mar '24 – Mar '25**
  - Developed a full-scale long-endurance aircraft for surveillance missions for Santa Clause, dubbed the *Silent Night Surveillant*.
  - Sized aircraft to meet mission-level requirements (e.g., endurance, climb rate, stall speed, payload mass) using first-principles flight mechanics.
  - Seeded a Monte Carlo optimization approach with the preliminary sizing to refine the aircraft design.
  - Modeled avionics and control system behavior in MATLAB and Simulink, validating performance against mission requirements.
  - Outlined the *Silent Night Surveillant* design process in a report, seen **here**.
  - Served as the lead launch vehicle and propulsion system engineer on a ten-person team in the development of the full-scale *Mayfly* spacecraft, designed to perform a sample retrieval mission.
  - Collaborated closely with the mission trajectory, power systems, and thermal systems engineers to ensure adherence to the mission timeline, sufficient power, and adequate heat dissipation on board the spacecraft.
  - Conducted trade studies comparing chemical and solar electric propulsion systems under mass and power constraints.
  - Iteratively evaluated propulsion systems to optimize for power, flight time, and mass.
  - Described the *Mayfly* design process in a report, seen **here**..
- **Hands-On Aircraft, Robotics & Controls Projects** **Mar '22 – Jun '24**
  - Designed, assembled, and refined RC quadcopter, RC airplane, model rocket, and a Mars rover designs across six different design projects.
  - Served as a system engineer and the lead electrical subsystem, guidance, navigation, and control system, and 3D design team engineer across multiple projects.
  - Integrated multiple subsystems in each project and validated designs against mission requirements.
  - Conducted research, collaborated with team members, and rigorously selected components, build prototypes, and iterated on each design.
  - Conducted experiments and simulations to evaluate materials and design variations, generating thrust, power consumption, and material strength profiles.

CONFERENCE PRESENTATIONS

- **SoCal Fluids XVIII**: *Development of a Flight Dynamics Model for D90* – April 19, 2025 | University of Southern California